Analysis of Variance in Pixel Intensities in Greyscale Bursts:

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Acknowledgments:

I hereby declare that several websites were consulted in course of this project for analysis, theory, specialized module functions and libraries and programs,a list of which has been provided in Appendix B.I am greatly indebted to our probability professor, Arnab Chakraborty for his timely help and invaluable suggestions on the analytical sections and methodology, that helped me to maintain a smooth progress. Though utmost care has been taken to keep the project error free, I apologise beforehand for any inadvertent error that might have crept in.

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1 Summary of the Project:

For this project, around 50 burst photos of a black text with large fonts on a white paper(preferably printed text) were taken. Then, using a specialized Python library, the photos were turned to grayscale mode. Next, variance of the pixel values(intensity) of individual pixels were computed using OpenCV package of python along with necessary analysis. The project focuses on whether the pixel values influence the variance and if yes, on the underlying reason behind it. Due to limited time and resources, some inevitable compromises have been made with the project. Here follows a few possible improvements on the project:

- The text was actually a handwritten font on white paper, shaded carefully with black sketch pens. Usage of printed text is advised to minimize chance errors and anomalous variance in pixel values.
- If printouts are used, make sure that the printer has adequate ink and that there are no ink-stains or unnecessary scratches, crumples on the printout/photocopy.
- A good camera should be used for more accurate results. If a mobile camera is used(like in this project), make additional considerations on outliers.
- Another idea is to verify the poisson statistics via plotting the histograms for specific pixel values and finding λ. For that purpose, a large number of photos must be clicked.

2 Objective of the Project:

We shall try to detect the regions with high variance and explain them. Intuitively, it seems sound that the boundary regions of the font will feature maximum variability which we shall check as a side-quest. The main goal of the project is to demonstrate that darker regions have more variability in terms of intensities or pixel values than the lighter ones and to provide explanation for these phenomena.

3 Tools Implemented:

The following tools were instrumental for the project.

- Python packages and modules
  - Numpy
  - Pandas
  - Matplotlib

  - Seaborn
  - os
  - opencv2

- 50 MP Sony AI primary sensor + 2 MP depth sensor mobile camera
- 50 burst photos of Pixel Gradient Stripes and the original 'Large Black Font on White paper' images

4 Methodology:

1. First 2 local folders containing the burst photos of the main image and the stripes gradient were created inside the python project directory of Visual Studio Code.(It is highly recommended to use the jupyter notebook extension in VSCode or the browser based Google Colab python notebooks for the analysis for quick reference instead of standard python scripts.)

2. Using the imread() and cvtColor functions we read all the images in grayscale mode and recorded their pixel values in a 2 dimensional numpy array(one for each copy).

3. To compute the mean intensity for individual pixels as well as the variance, we first stacked the 2d numpy arrays into a 3d array and computed the mean of pixel values and variance of pixel values along the stacking axis(by default axis=0).This operations returned a 2d array image, which we shall call the ***Representative Image***.

4. The intensity of the ***Representative Image*** is stored in an array called **mean_pixel_value**, while the variance is stored in **variance_pixel_value**. Both these arrays have dimensions $4096 * 3072$.

5. Then a heatmap was plotted to detect the regions with high variance of pixel values. As the boundary regions had quite a bit higher variance than the rest of the image, the heatmap was masked to highlight the variance disparity between the shaded darker region and the lighter background.

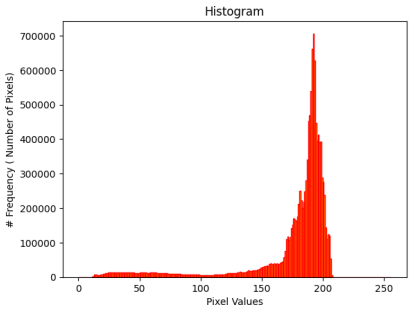
6. Numerical and graphical analysis was done to confirm that on an average, the darker regions had accounted for more variance than the lighter regions.

7. Finally we used the Theory of Charge Coupled Devices to explain the phenomena, keeping into account the shortcomings of the camera, procedure of the project and outliers arising from chance outliers.

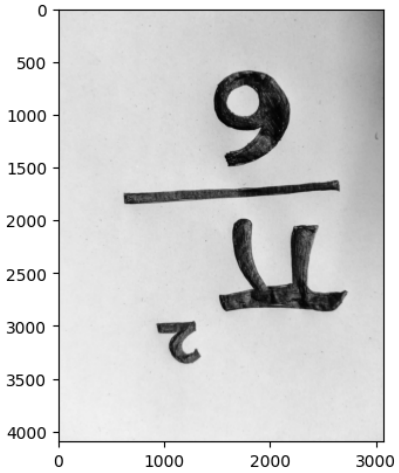
5 Analysis and Observations:

5.1 Image and Differentiating the two Clusters:

First of all, we shall plot a histogram of the mean pixel values of the greyscale images to identify the dark and light regions(approximately). We are essentially taking our ***Representative Image*** and plotting a histogram of the pixel intensities .



(a) Pixel Intensities v/s # Pixels



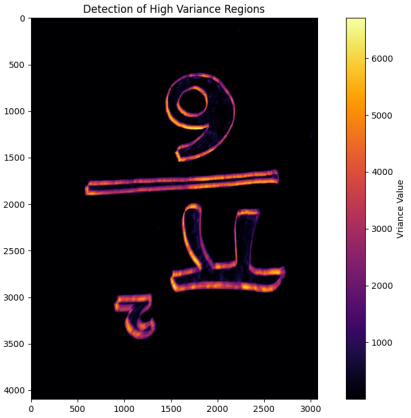
(b) Sample Greyscale Image

Figure 1: Differentiating Light and Dark Regions

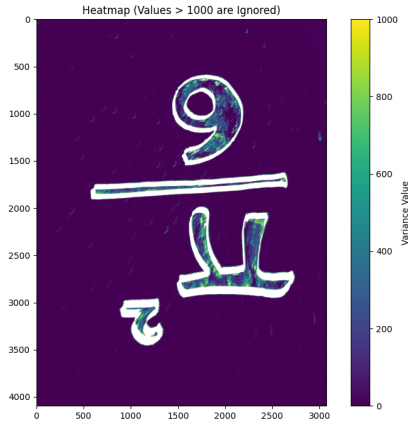
It is evident from the figure that there are 2 separate clusters of pixels in my image: the darker pixels having a peak before 100 and the lighter ones around 200(as expected, the number of whitish pixels outnumber the blackish pixels manifold). There is a prominent valley in between from where we can choose the critical region.
In this project, I have taken pixel values ≤ 125 to be black and pixel values ≥ 170 to be white.

5.2 Variance Heatmaps of the Image:

Now we shall plot 2 heatmaps to detect the higher variance regions within the ***Representative Image***.



(a) Heatmap of Variance



(b) Masked Heatmap to compare variance in intensities of the lighter and darker regions

Figure 2: Variance Heatmaps

As expected, in the unfiltered heatmap, we can observe greater variance in the boundary regions in comparison to the rest. As the extremely high variance in the boundaries almost muffle the variance difference between the dark shaded region and the white background, we plot another masked heatmap that ignores all the variance greater than a certain value(marking those regions by white). The second figure highlights that the shaded regions have more pixel value variance than the lighter background.

5.3 Numerical Evidence:

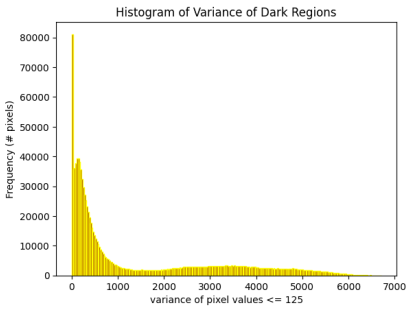
The numerical results, calculated using python, show the average, minimum and maximum variance for the 'Black' and 'White' regions.

Table 1: Numerical Results

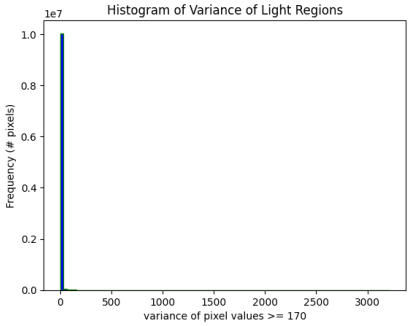
Variance Results	Black Region	White Region
Minimum	0.84	0.39
Maximum	6706.58	3214.69
Mean	1432.26	12.65

5.4 Variance Histograms:

Finally we shall plot 3 histograms: 2 regular ones that plot variance and the frequency for the two lots and one 2 dimensional hexagonal histogram plotting variance(y) against pixel values(x). As the plot suggests, most of the light pixels



(a) Variance Histogram for the Blackish Regions



(b) Variance Histogram for the Whitish Regions

Figure 3: Regular Histograms of Variance v/s Frequency

show almost no variance in pixel intensities, while the dark pixels feature quite a bit of variability. It must be noted that the numbers of light pixels outnumbering dark pixels by several powers of 10, our analysis must be relative(normalized by the total numbers in each lot)- which is done in the histograms.
This phenomenon is better visualized using a 2 dimensional hexagonal-bin histogram that plots variance against pixel values with each hexagonal cell coloured according to a colourbar depending on the frequency of pixels lying there. The

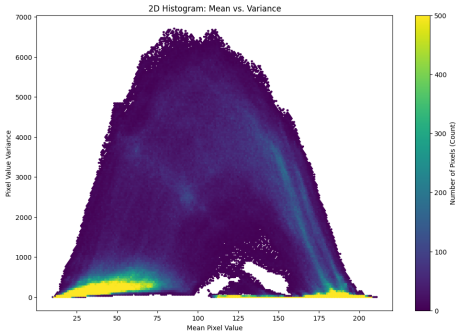


Figure 4: 2D Histogram plotting Variance against pixel intensity

bright yellow patches on both side of the histogram along with the peaks being to the left of 125 mark suggests that on the whole darker regions have more variance and also justifies the results given in the table.

5.5 Important Observations:

We conclude this section by providing some important takeaways from our analysis.

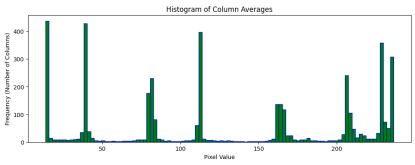
- The boundary regions of the black fonts against the white theme show extremely high variability of pixel intensities.
- Removing the effect of the borderline variance, we can see that the darker shaded regions have more variance of pixel values than the lighter background.

Our main goal is to explain these phenomena throughout the next section.

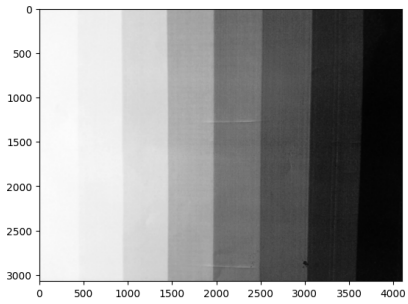
6 Stripes Analysis:

6.1 Variance Results:

To work towards establishing the results, we shall first take a photo of 8 adjacent stripes, with average pixel values varying from 0 to 256 in a somewhat regular manner, for our auxiliary analysis. Next, we snap 50 burst shots of the photo and analyze them a similar manner. Taking a mean of the greyscale versions of the 50 images, we form a representative image here as well. We first make sure that all the 8 stripes are identified by plotting a histogram of the mean pixel values. As there will be significant jumps in the consecutive stripes in terms of intensity, we can expect to have 8 peaks and valleys in between the peaks. As



(a) Histogram of Mean Pixel Values



(b) A Greyscale Sample of the Stripes

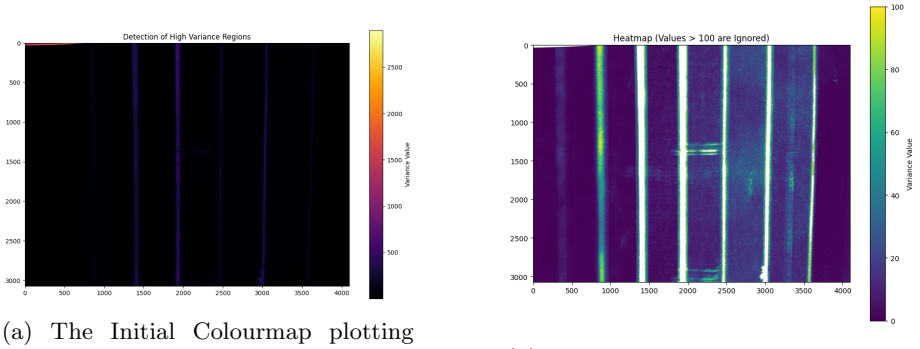
Figure 5: The Stripes Bursts

evident from the figure 5(a), all the stripes have been recognized by the sensor. Then we can take the values ≤ 130 as blackish shades and the mean intensities ≥ 200 as blackish shades. Proceeding as before produces the results:

Table 2: Numerical Results

Variance Results	Black Region	White Region
Minimum	0.28	0.15
Maximum	638.74	2621.08
Mean	30.30	12.12

We now simply plot a colourmap to detect the high variance regions. The first image displays that the boundary regionms have relatively high variance. Then we again mask out the high variance regions by restricting the colourbar and note the variance disparity between the stripes.



(a) The Initial Colourmap plotting Variance

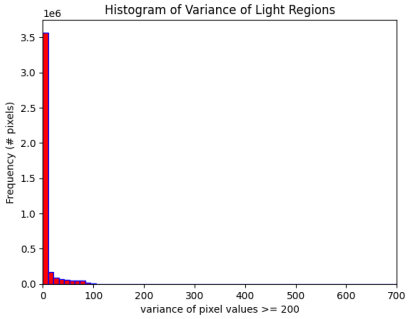
(b) The Effect of Boundaries have been Removed

Figure 6: Variance Colourmaps

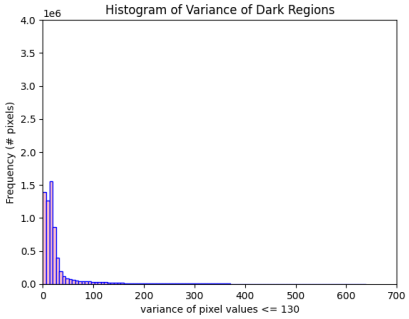
We shall also note a very interesting thing here. There is abnormal variance in the light pixels. Another heatmap, that ignores all the variance values ≤ 1000 actually reveals that the upper left corner of the stripes 'Representative Image' can account for the abnormally high variances. Thus, it is reasonable to consider these few pixels as outliers and ignore them for general analysis.

6.2 variance Histograms:

Just like the main image, the variance v/s frequency histogram plots for the 'Dark' and 'Light' regions are attached here. For easy comparison, the variance axis of the light regions have been trimmed to 700 units, while the frequency axis of the Dark regions have been stretched to $4 * 10^6$ pixels. Again, a large number of light pixels hardly have any variance. Also, those showing a little bit variance are largely outnumbered by the corresponding dark pixels.



(a) Histogram of variance of pixel intensities in the Light Regions

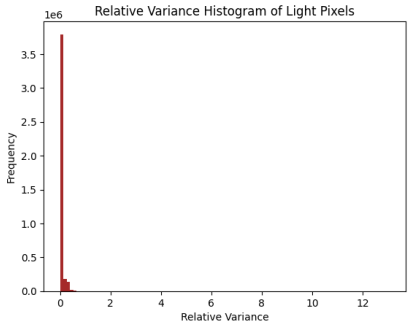


(b) Histogram of variance of pixel intensities in the Dark Regions

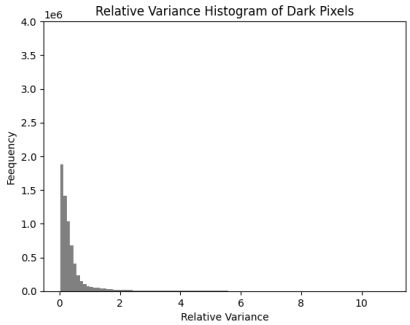
Figure 7: variance Histograms of the Stripes

6.3 Relative Variance:

Relative Pixel Variance is essentially pixel variance normalized by pixel intensity. Checking Relative Variance for the Dark and Light regions show similar results, as given below:



(a) Histogram of relative variance of pixel intensities in the Light Regions



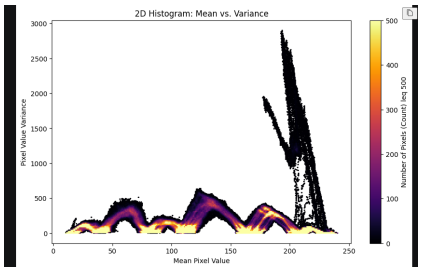
(b) Histogram of relative variance of pixel intensities in the Dark Regions

Figure 8: Relative variance Histograms of the Stripes

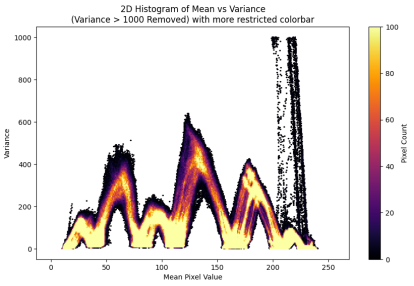
6.4 2D Histogram of Variance v/s Pixel Values:

The 2 dimensional histogram for pixel values and variances give a wonderful insight in this case.
We can easily observe the following things from the diagram:

- There are separate bump like structures that peaks around the mean intensities of the stripes.



(a) Original 2D Histogram for the Stripes



(b) Masked Histogram

Figure 9: 2D Histogram visualizing Variance v/s Pixel Value

- Some of the white pixels have extremely high variance.
- Such pixels are very few in number, as indicated by the adjacent colourbar.

7 Explanations:

7.1 Working Principals:

The camera used in modern smartphones actually use complementary metal oxide semiconductor sensors for image processing. Light (in form of photons) reflect from the surface of the subject and pass through the camera lens to hit the pixels. The pixels use photo diodes to generate electrons from the incident photons($N_e \propto N_{ph}$). Then the accumulated charge is converted to a proportional voltage($V = Q/C$) using capacitors.

7.2 Reasons for Variances:

There are 3 major sources of noise or variance in intensities; namely shot noise, dark noise and read noise.

- **SHOT NOISE:**The number of photons collected in a pixel during the exposure time follows a Poisson statistics,(with mean almost proportional to the values derived in the mean_pixel_value matrix) giving rise to a natural source of variance in the pixel values.For poisson, variance equals mean. Hence $\sigma^2_{shot} = N_{ph}$.
- **DARK NOISE:**Even when there is no exposure of light, the photodiodes keep releasing electrons(dark current), the number of which again follows a poisson statistics with mean N_d.Once accumulated in the pixel, they are indistinguishable from the photoelectrons, hence contributing to the variance. $\sigma^2_{dark} = N_d$.

- **READ NOISE:**The obvious source of variance in any physical experiment from defective tools, faulty measuring procedures, the specific imaging technology implemented etc.

All 3 sources being mutually independent, the total variance in pixel intensities should be:

$$\sigma^2 = \sigma^2_{shot} + \sigma^2_{dark} + \sigma^2_{read} = N_{ph} + N_d + \sigma^2_{read}$$

7.3 Expectation:

The lighter a region is, the more is its pixel value(number of electrons generated and collected in the pixel well during the exposure time,by both photons and dark current). For the brighter regions in an image, N_{ph} outnumbers dark electrons manifold and read noises are also insignificant. Hence, Variance $\sigma^2 \approx N_{ph}$. therefore, we should expect to observe more variance in the lighter background of our image.
The SNR (signal to noise ratio, $\propto \sqrt{N_{ph}}$), another measure of variability should also be high in the brighter regions.
The relative variance(i.e. variance normalized by mean pixel intensity, $\propto \frac{1}{\sqrt{N_{ph}}}$) should be more in the darker regions.

7.4 Conclusion from the Observations:

1. Unlike the theory, the project produces more absolute variance in the darker regions on average. Though this can result from technical mistakes during the project, faulty devices some sources suggest that sometimes in dark regions read noise dominates shot noise leading to the observed phenomenon.
   Long exposure during the burst clicks can also result in generation of thermal electrons, thereby accumulating dark noise.

2. The relative variance scenario is portrayed perfectly in the project.

3. The interesting patterns featured in figure 9 can be explained by Fixed Pattern Noise (Photo response non uniformity), where homogeneous pixel values accumulate around a certain mean, precisely the ones displayed in figure 5(a).

4. The heatmap in figure 6 (b) shows that the black stripes (except the rightmost one) feature larger variability.

There were a few observations that I was unable to explain. According to my judgment they were due to methodological errors or device limitations.

8 Appendix A: Python Codes

8.1 Importing Libraries and Converting Images into Greyscale:

```
import os
import numpy as np
import cv2
import matplotlib.pyplot as plt
subfolder_path='./burst_photos'
files = [f for f in os.listdir(subfolder_path) if f.endswith((
    '.jpg'))]
pixel_data_list=[]
greyscale_images=[]
for file_name in files:
    file_path=os.path.join(subfolder_path, file_name)
    image=cv2.imread(file_path)
    grey_photo=cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    greyscale_images.append(grey_photo)
    img_grey=cv2.imread(file_path,0)  #Reading the images in
        greyscale mode
    pixel_data_list.append(img_grey)
    print(f"Processed: {file_name}, Shape: {img_grey.shape}")
    #Thus created 50 numpy arrays of pixelvalues
```

8.2 Fundamental Analysis:

```
greyscale_images_stack= np.stack(greyscale_images, axis=0)
# All the analysis will produce a single 4096*3072 numpy array
    of a representative photo
mean_pixel_val=np.mean(greyscale_images_stack, axis=0)
variance_pixel_value=np.var(greyscale_images_stack, axis=0)
vmin=np.min(variance_pixel_value)
vmax=np.max(variance_pixel_value)
print(vmin)
print(vmax)
vmean=np.mean(variance_pixel_value)
print(vmean)
```

8.3 Detection of 'Dark' and 'Light' Regions:

```
plt.hist(mean_pixel_val.ravel(), bins=256, range=[0,256], color
    ='orange', edgecolor='red')
plt.title('Histogram Burst Photos')
plt.xlabel('Pixel Values')
plt.ylabel('# Frequency ( Number of Pixels)')
plt.show()
# This results in the histogram of Figure 1 (a)
```

8.4 Detection of High Variance Regions(via Heatmap):

```
plt.figure(figsize=(12,8))
plt.imshow(variance_pixel_value, cmap='inferno', vmin=vmin,
    vmax=vmax)
plt.colorbar(label='Variance Value')
plt.title("Detection of High Variance Regions")
plt.show()
#This code gives rise to the heatmap of Figure 2(a)

# Announcing a threshold variance for mask
threshold=1000
# 2. Create a masked array where values > threshold are ignored
masked_variance_pixel_value = np.ma.masked_where(
    variance_pixel_value > threshold, variance_pixel_value)
plt.figure(figsize=(10, 8))
plt.imshow(masked_variance_pixel_value, cmap="viridis", vmin=0,
     vmax=threshold)
plt.colorbar(label='Variance Value')
plt.title(f"Heatmap (Values > {threshold} are Ignored)")
plt.show()
#This code gives rise to the one in Figure 2 (b), that compares
     the variance of the darker and the lighter regions
    ignoring the boundary regions
```

8.5 Variance Comparison:

```
#Accessing the Coordinates with pixel value >170
coordinates_geq170= np.argwhere(mean_pixel_val > 170)
# Calculating the variance of the whitish shades
variance_pixel_value_geq170=[]
for i in coordinates_geq170:
    variance_pixel_value_geq170.append(variance_pixel_value[i
        [0], i[1]])

print(np.mean(variance_pixel_value_geq170))
print(np.min(variance_pixel_value_geq170))
print(np.max(variance_pixel_value_geq170))

#Accessing the Coordinates with pixel value <125
coordinates_leq125= np.argwhere(mean_pixel_val <125)
# Calculating the variance of the whitish shades
variance_pixel_value_leq125=[]
for i in coordinates_leq125:
    variance_pixel_value_leq125.append(variance_pixel_value[i
        [0], i[1]])

print(np.mean(variance_pixel_value_leq125))
print(np.min(variance_pixel_value_leq125))
print(np.max(variance_pixel_value_leq125))

#The aforementioned codes gave rise to the Table 1.
#Here follows the corresponding Histograms:

plt.hist(variance_pixel_value_geq170, bins=100, color='blue',
    edgecolor='green')
```

```
plt.xlabel('variance of pixel values >= 170')
plt.ylabel('Frequency (# pixels)')
plt.title('Histogram of Variance of Light Regions')
plt.show()

plt.hist(variance_pixel_value_leq125, bins=250, color='maroon',
     edgecolor='yellow')
plt.xlabel('variance of pixel values <= 125')
plt.ylabel('Frequency (# pixels)')
plt.title('Histogram of Variance of Dark Regions')
plt.show()

# These codes gave rise to the histograms in Figure 3.
```

8.6 2 Dimensional Hexbin Histogram: Mean Intensity v/s Variance with Colourbar

```
x = mean_pixel_val.flatten()
y = variance_pixel_value.flatten()
# 3. Create the 2D Histogram / Hexbin Plot
plt.figure(figsize=(12, 8))
hb = plt.hexbin(x, y, gridsize=200, cmap='viridis', mincnt=1,
    vmin=0, vmax=500)
plt.colorbar(hb, label='Number of Pixels (Count)')
plt.xlabel('Mean Pixel Value')
plt.ylabel('Pixel Value Variance')
plt.title('2D Histogram: Mean vs. Variance')
plt.show()

# The 2D histogram of figure 4 is the product of this code.
```

8.7 Poisson Statistics

This code basically explores the underlying statistics of pixel intensities for a specific pixel across the bursts. The peaks in the histogram usually determines the λ for that pixel. However, the sample size being too small, these findings were not included in the core project due to vague inference.

```
    pixel_series=greyscale_images_stack[:, 1000,2000]
    # Here the pixel under scrutiny is (1000, 2000)
    plt.hist(pixel_series, bins=20, color='red', edgecolor='
        black')
    plt.title('Pixel Value Distribution across 50 Frames')
    plt.xlabel('Pixel Intensity')
    plt.ylabel('Frequency')
    plt.show()
```

These were the codes used to analyze the 50 burst photos. Codes used for the 'Stripes' photos for explanation being more or less similar, are ignored here.Some other explanatory codes used for the 'Stripes' exclusively are attached next.

8.8 Variance and Relative Variance Histograms

```
relative_variance_leq130= [x/y for x, y in zip(
    variance_pixel_value_leq130, mean_pixel_value_leq130)]
plt.hist(relative_variance_leq130, bins=100, color='grey')
plt.ylim(0, 4000000)
plt.title('Relative Variance Histogram of Dark Pixels')
plt.xlabel('Relative Variance')
plt.ylabel('Feequency')
plt.show()

plt.hist(variance_pixel_value_geq200, bins=250, color='red',
    edgecolor='blue')
plt.xlim(0, 700)  # To compare properly at the same scale with
    the darker regions, higher variances are ignored.
plt.xlabel('variance of pixel values >= 200')
plt.ylabel('Frequency (# pixels)')
plt.title('Histogram of Variance of Light Regions')
plt.show()
# Bin size anomaly is to divide the x range into intervals of
    almost same length as that of dark pixels
```

8.9 Detecting and Filtering Outliers

Technically it is same as the masking used earlier but the objective is a little different.

```
threshold_2=1000
# 3. Create the Mask: Identify regions where variance >
    threshold
# This returns a boolean array where True is high variance
high_var_mask = variance_pixel_value > threshold_2

# 4. Apply mask to the mean map (or original image)
# Option A: Create a new image keeping only high variance areas
high_var_regions = np.where(high_var_mask, mean_pixel_value, 0)

# Option B: Get coordinates (y, x) of high variance pixels
high_var_pixels = np.argwhere(high_var_mask)
print(f"Number of high-variance pixels: {len(high_var_pixels)}"
    )

# 5. Visualization
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.title("Variance Map")
plt.imshow(variance_pixel_value, cmap='viridis')
plt.colorbar()

plt.subplot(1, 3, 2)
plt.title("High-Var Mask (>1000)")
plt.imshow(high_var_mask, cmap='gray')

plt.subplot(1, 3, 3)
plt.title("Filtered Regions")
```

```
    plt.imshow(high_var_regions, cmap='gray')
    plt.show()
```

```
    mask = variance_pixel_value <= threshold_2

    # Apply mask to keep only valid data points
    filtered_mean = mean_pixel_value[mask]
    filtered_variance = variance_pixel_value[mask]

    print(f"Original shape: {mean_pixel_value.shape}")
    print(f"Filtered points: {filtered_mean.shape}")
```

9 Appendix B: References

The following websites and online resources have been consulted time and again in course of the project:

- Greyscale in OpenCV: This Geeks for Geeks article was used for conversion and handling of the images using the opencv2 library in python.
- Scientific Image Processing: Articles under the topic Scientific cameras, especially C10) to C13) inclusive, were consulted for the explanation section.
- Several Stack Overflow and overleaf forums turned out handy for fixing both the python and LaTEX codes.
- Thorlabs Fixed Pattern Noise